University of Barishal

Department of CSE

B.S (Hon's) 3rd Year 2nd Semester Final Examination- 2024

Exam Session: 2021-22

Course name: Mathematical Analysis for Computer Science **Course code: CSE 3201**

Total Marks: 60 **Time: 3 Hours**

Answer any FIVE questions. Marks are indicated at the end of the question

Q1. (a) Define and explain different types of random variable. [4]

(b) Write down the properties of random variable. [4]

(c) Write down the differences between general variable and random variable. [4]

Q2. (a) What do you mean by stochastic process? Briefly describe different types of stochastic process with example. [4]

(b) What is second order process? Write down the properties of covariance function. [4]

(c) Consider the process $X(t) = A_1 + A_2 t$ where A_1 *and* A_2 are independent random variables with $E(A_i) = a_i, V(A_i) = \sigma_i^2$; $i = 1, 2$. Test the stationarity. [4]

Q3. (a) Consider the process $[X(t), t \in T]$ whose probability distribution, under a certain condition, is given by [6]

$$\Pr[X(t) = n] = \frac{(at)^{n-1}}{(1+at)^{n+1}} \quad , \qquad n = 1, 2, \ldots \ldots$$

$$= \frac{at}{1+at} \quad , \qquad n = 0$$

Test the stationarity.

(b) Define martingales process. Let $\{Z_i\}$; $i = 1, 2, \cdots$ be a sequence of *i.i.d.* random variables with mean 0 and let $X_n = \sum_{i=1}^{n} Z_i$, then show that $\{X_n\}_{n=1}^{\infty}$ is a martingale process with respect to $\{Z_n\}_{n=1}^{\infty}$ [6]

Q4. (a) Define the following terms: (i) Continuous time markov chain (ii) Pure Birth Process (iii) A Birth Process with Linear Birth Rate. [3]

(b) Determine the Probability Function for the Yule Process. [5]

(c) State and prove Chapman Kolmogorov's Forward differential equations. [4]

Q5. (a) Define transition probability. State and derive the Chapman Kolmogorov equation for determining the higher order transition probabilities of a Markov Chain. [5]

(b) Describe the classification of states according to communication. [3]

(c) Show that, for any state j $\sum_{n=0}^{\infty} P_{jj}^{(n)} = \frac{1}{1 - F_{jj}}$ and if state j is recurrent then $\sum_{n=0}^{\infty} P_{jj}^{(n)} = \infty$ [4]

Q6. (a) Define the following terms: i) Renewal process, ii) Renewal function, iii) Renewal equation. 3

(b) State and derive the renewal process. Show that the renewal function also satisfies the equation $M(t) = F(t) + \int_0^t m(t-x)dF(x)$. 6

(c) Suppose we have a renewal process whose mean-value function is given by $m(t) = 2t; t \geq 0$ 3

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a) What is the probability that 2 or 3 renewals occur by time 10

b) What is the probability that at least 1 renewals occurs by time 10

c) What is the distribution of number of renewals by time 10

Q7. (a) Under usual notations, prove that [4]

$$\lim_{h\to 0} \frac{1 - P_{ii}(h)}{h} = v_i$$

$$\lim_{h\to 0} \frac{P_{ij}(h)}{h} = q_{ij} \qquad for\ i \neq j$$

(b) Show that, the average amount of time that a customer spends for waiting in the system is [4]

$$W = \frac{1}{(\mu - \lambda)} = \frac{1}{\mu(1-\rho)}$$

(c) Suppose in a super market there is only one cashier. Customers arrive at random at a rate of 12 per hours and the service time has negative exponential distribution. At what rate should a customer work to ensure that at most 90% of customers do not have to wait for more than 4 minutes? [4]

Q8. (a) What do you mean by M/M/1 Queuing Model? Write down the properties of M/M/1 model. [5]

(b) Describe the Service mechanism or service facility of queuing process. [7]

(c) A transport company has a single unloading berth with vehicles arriving in a Poisson fashion at an average rate of three per day. The unloading time distribution for a vehicle with 'n' unloading workers are found to be exponentially with an average unloading time (1/2)×n days. The company has a large labor supply without regular working hours, and to avoid long waiting lines, the company has a policy of using as many unloading groups of workers in a vehicle as there are vehicles waiting in line or being unloaded. Under these conditions find

(i) What will be the average number of unloading group of workers working at any time?

(ii) What is the probability that more than 4 groups of workers are needed?

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